Fresh Produce Supply Chain: Performance Review & Action Plan

Demand forecasting, inventory optimisation and waste reduction for 5 products (Tomato, Onion, Potato, Cucumber, Capsicum)

Bottom line. We never ran out of stock (fill rate 100%, 0 stockout days), but we paid for it with over-stocking: **20.7% of everything we sold was written off (1,755 kg, about ₹49,554 at assumed prices)**. We hold about **1,670 kg** of stock on average against a need of about **720 kg**. Tightening reorder points and fixing expiry control can cut waste by half and save roughly **₹24,777 per month** without hurting availability.

TOTAL DEMAND	FORECAST, NEXT 7 DAYS	FILL RATE
8,479 kg in 30 days	**2,000** kg expected demand	**100%** demand fully served
TOTAL WASTE	**WASTE % OF DEMAND**	**EXCESS STOCK**
1,755 kg written off	**20.7%** too high: main problem	**950** kg above target (avg)

The three business questions, answered

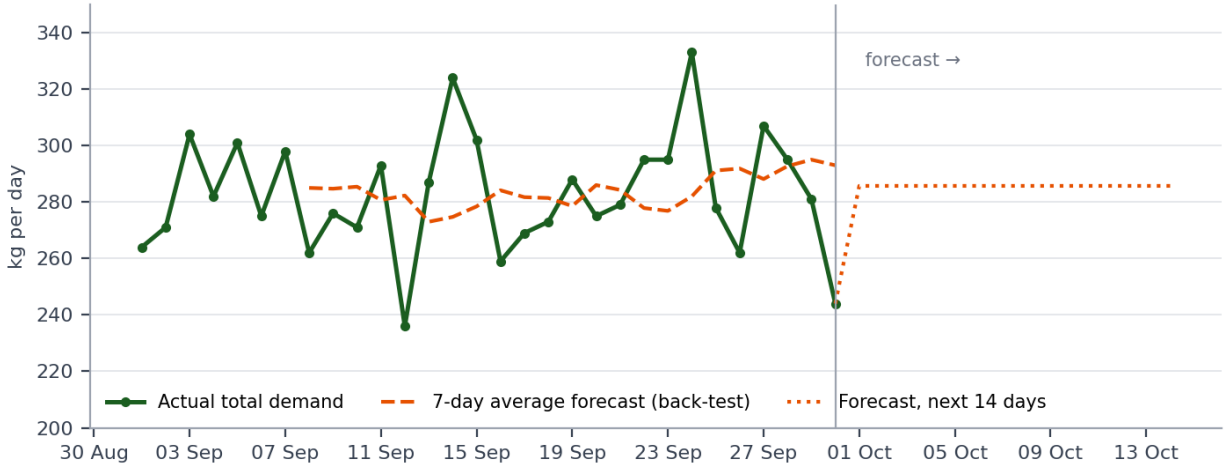
1. What to order?	Plan for about **2,000 kg** over the next 7 days. Potato needs the most (546 kg), then Onion (503 kg) and Tomato (449 kg). Per 4-day order cycle, order about **1,160 kg** in total (quantities per product on page 3). The 7-day average forecast is 92.6% accurate for total demand and 81–86% for individual products.
2. When to order?	Order when stock falls to the reorder point. On 30 Sep, **Capsicum is below its reorder point: order 130 kg now**. Cucumber (3.3 days of cover) is next, followed by Tomato, Onion and Potato over the following 2–4 days.
3. How to cut waste?	Waste is 20.7% of demand while fill rate is 100%, so the problem is too much stock, not too little. Use first-expiry-first-out (FEFO), order smaller batches more often, and divert near-expiry stock to cut/chopped products or quick sales. A 50% reduction saves about 878 kg (≈ ₹24,777) per month.

Decisions requested from you

- **Approve** the new reorder points (total 1,100 kg instead of the current 1,219 kg) and lower safety stock (140 kg instead of 259 kg), reviewed monthly.
- **Mandate FEFO and lot tracking** (received date and expiry date for every lot) from this week, so we can find the real cause of write-offs.
- **Agree a pilot** of smaller, more frequent orders for Tomato, Cucumber and Capsicum, our fastest-spoiling items.

1 · Demand and forecast

Total demand was **8,479 kg** over 30 days (average 283 kg per day, range 236–333 kg). Potato and Onion are 54% of all demand. Because daily demand swings by about ±17% around the average, a fixed order size is risky; we should order from a rolling forecast.

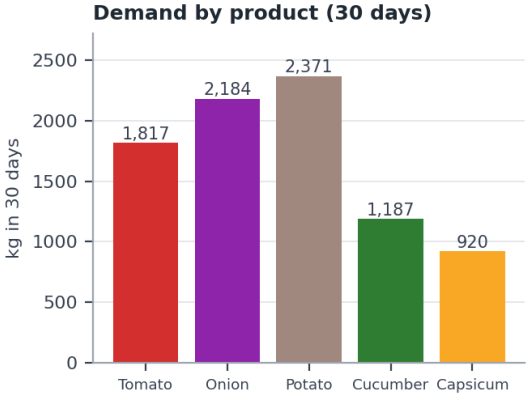


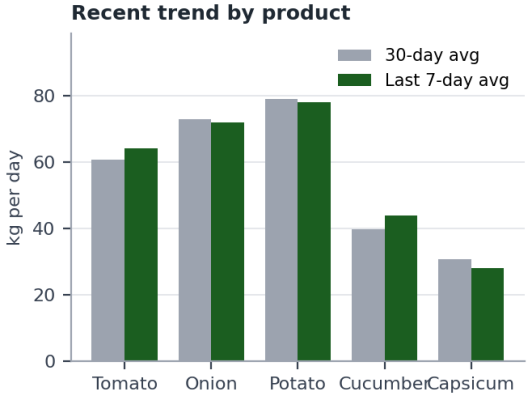
Forecast method: average of the previous 7 days. Accuracy tested on days 8–30. Future days use the latest 7-day average (flat line).

Product	Avg demand / day (kg)	Last 7-day avg (kg)	Trend vs 30-day avg	Forecast 7 days (kg)	Forecast 14 days (kg)	Peak day (kg)	Forecast accuracy
Tomato	60.6	64.1	+5.9%	449	898	76	83%
Onion	72.8	71.9	-1.3%	503	1,006	91	86%
Potato	79.0	78.0	-1.3%	546	1,092	104	81%
Cucumber	39.6	43.9	+10.8%	307	614	50	85%
Capsicum	30.7	27.9	-9.2%	195	390	39	83%
Total	**282.6**	**285.7**	**+1.1%**	**2,000**	**4,000**	**333**	**93%**

What this means

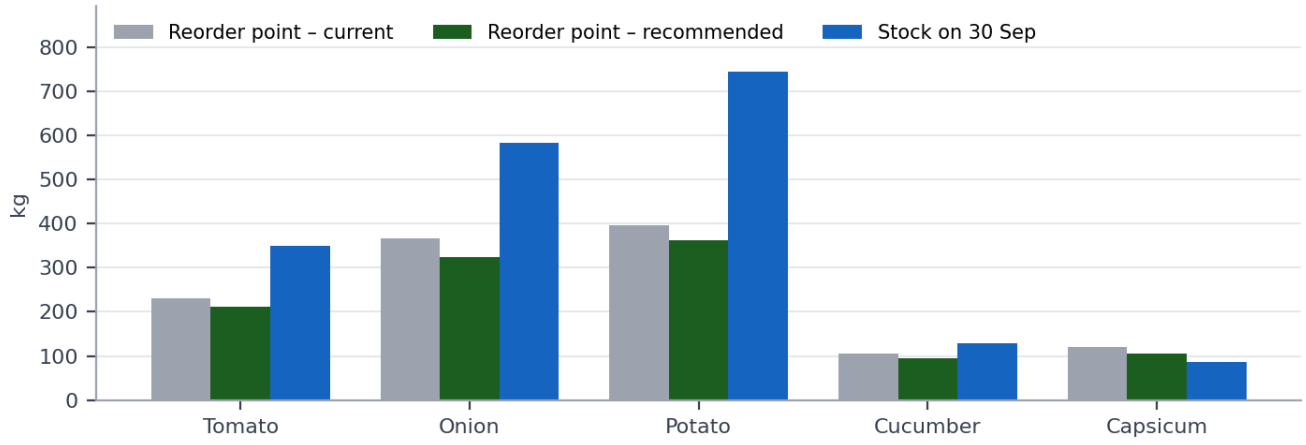
- **Priority products:** Potato (546 kg) and Onion (503 kg) drive the forecast. They also have the longest lead time (4 days), so ordering late hurts most here.
- **Rising demand:** Cucumber is up 11% and Tomato up 6% in the last 7 days compared with the 30-day average. Capsicum is down 9%.
- **Forecast quality:** good enough to order from (total 93%, products 81–86%). Product-level error is higher because individual products are noisier. Weekday, festival and weather effects can be added once more history is available.





2 · Inventory, safety stock and reorder points

Safety stock = Z × demand variability × √lead time (95% service level). Reorder point = average daily demand × lead time + safety stock. The recommended reorder points total **1,100 kg** versus **1,219 kg** today. Current safety stock (259 kg) is about **1.8×** what the statistics require (140 kg).

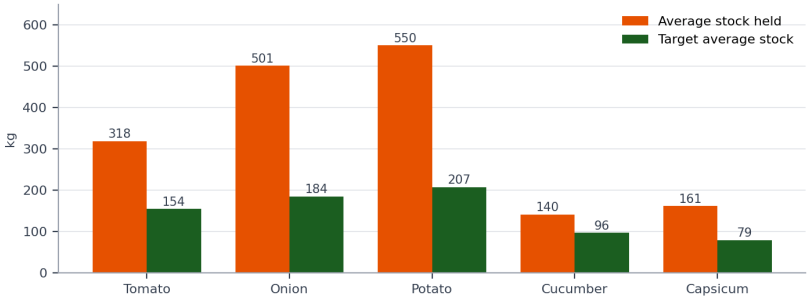


Product	Lead time (d)	Safety stock rec. (kg)	Reorder pt rec. (kg)	Reorder pt current (kg)	Stock 30 Sep (kg)	Days of cover	Status	Order qty (kg)	Next order date
Tomato	3	29	211	231	349	5.8	OK	250	02 Oct
Onion	4	34	325	366	584	8.0	OK	300	03 Oct
Potato	4	47	363	397	745	9.4	OK	320	04 Oct
Cucumber	2	16	95	105	129	3.3	OK	160	30 Sep
Capsicum	3	14	106	120	87	2.8	ORDER NOW	130	30 Sep
Total		**140**	**1,100**	**1,219**	**1,894**	**6.7**		**1,160**	

Status: ORDER NOW = below reorder point; WATCH = within 20% above it; OK = otherwise. Order qty = average daily demand × 4-day order cycle, rounded up to the next 10 kg.

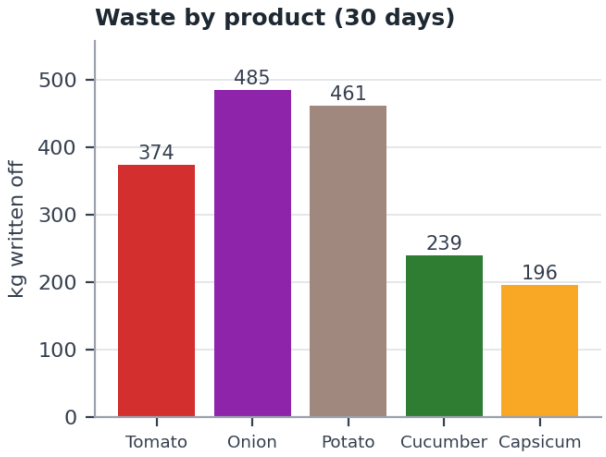
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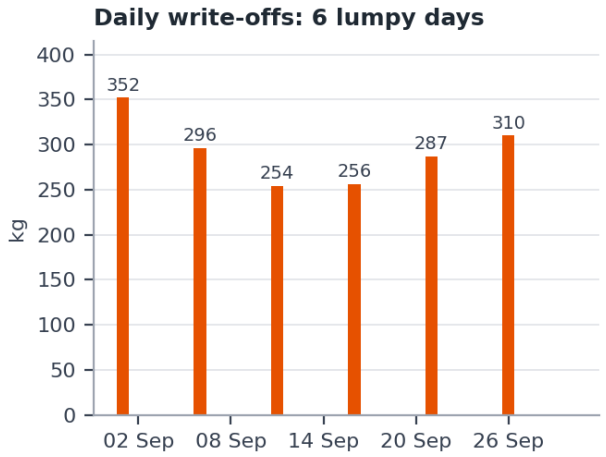
- **Act today:** Capsicum has 87 kg against a reorder point of 106 kg (2.8 days of cover). Order 130 kg. Cucumber has 3.3 days of cover and its order is due within a day.
- **Too much buffer elsewhere:** on average we hold 1,670 kg against a target of 720 kg, about 950 kg extra. Potato (343 kg) and Onion (317 kg) account for most of it.
- **The 100% fill rate is not free:** every product-day with stock above target is stock that can expire. Lowering buffers should keep fill rate high; track it weekly to confirm. In September, 32 of 150 product-days were below the current reorder point.



3 · Waste and expiry

1,755 kg was written off, equal to **20.7%** of demand and about **₹49,554** at assumed buying prices. Every product is in the 19–22% range, so this is a system-level issue (stock control, ordering and storage), not one bad product.





Product	Demand (kg)	Waste (kg)	Waste % of demand	Share of total waste	Assumed price (₹/kg)	Waste value (₹)	Saving at 50% (kg)	Saving at 50% (₹)
Tomato	1,817	374	20.6%	21%	25	9,350	187	4,675
Onion	2,184	485	22.2%	28%	30	14,550	242	7,275
Potato	2,371	461	19.4%	26%	22	10,142	230	5,071
Cucumber	1,187	239	20.1%	14%	28	6,692	120	3,346
Capsicum	920	196	21.3%	11%	45	8,820	98	4,410
Total	**8,479**	**1,755**	**20.7%**	**100%**		**49,554**	**878**	**24,777**

Prices are assumptions for illustration. Replace with real buying rates to get a true rupee figure.

What this means

- **Waste comes in lumps:** it was recorded on only 6 of 30 days, up to 352 kg at once. This looks like old stock being written off in batches, which means we find out late.
- **Where the money is:** Onion is the biggest loss (485 kg, ₹14,550). Tomato, Cucumber and Capsicum together lose 809 kg (46% of all waste) and are the most perishable, so they respond best to smaller, more frequent orders.
- **Prize:** cutting waste by 50% saves about 878 kg (≈ ₹24,777) per month at assumed prices.

4 · Action plan

Proposed owners and timing are suggestions for you to confirm.

#	Action	Proposed owner	When	Expected benefit
1	**Place today's orders:** Capsicum 130 kg now; Cucumber 160 kg within a day; then Tomato 250, Onion 300, Potato 320 kg as they hit reorder point.	Procurement	This week	Avoids stockouts on the lowest-cover items.
2	**Adopt the recalculated reorder points and safety stock** (1,100 kg / 140 kg total). Review monthly; start with Onion and Potato (4-day lead time).	Operations + Procurement	Next 2 weeks	Lower stock held without risking stockouts.
3	**Order from the 7-day rolling forecast.** Refresh weekly. Use the Stock status column as a daily buying alert.	Procurement	Weekly routine	Right quantity ordered; fewer emergency purchases.
4	**Enforce FEFO and lot tracking:** received date and expiry date on every lot; daily stock-age check.	Warehouse	Start this week	Finds the root cause of waste; stops lump write-offs.
5	**Pilot smaller, more frequent orders** for Tomato, Cucumber, Capsicum. Keep safety stock; remove the extra buffer.	Procurement + Warehouse	30-day pilot	Part of the ≈ 878 kg / ₹24,777 monthly saving.
6	**Divert near-expiry stock** to cut/chopped, ready-to-use products or quick sales to HoReCa customers.	Sales + Operations	Next 30 days	Turns waste into sales.
7	**Replace assumptions with real data:** shelf life, supplier on-time %, price per kg, customer-wise orders. Recalculate service level, costs and savings.	Analytics	Next 60 days	Moves the model from practice to real decisions.

How to read the numbers (assumptions and limits)

- **Data is simulated:** 30 days (1–30 Sep 2026), 5 products. Results show the method, not real business performance.
- **Missing inputs:** shelf life, buying price, supplier on-time delivery and customer orders are not available. Rupee values use assumed prices (Tomato 25, Onion 30, Potato 22, Cucumber 28, Capsicum 45 ₹/kg).
- **Method:** forecast = 7-day moving average · safety stock = Z × standard deviation × √lead time (95% service level) · reorder point = average demand × lead time + safety stock · target average stock = safety stock + half an order.
- **Small sample, no seasonality:** thirty days cannot capture festivals, weather or price changes. Re-test the reorder points on 3+ months of real data before changing live buying rules.

Supporting file: Fresh_Produce_Supply_Chain_Analytics.xlsx. Every number above comes from that workbook (Dashboard, Demand, Inventory, Waste, Recommendations, Data, ReadMe), where the calculations can be traced cell by cell.